ECE 313: Probability with Engineering Applications

2025 Fall   Instructors: Piao Chen & Xu Chen

Homework 11

Name: ______________________
Student ID: ______________________

Due Date: December 26 23:59, 2025

Problem 1. Let $X_1, X_2, \ldots$ be a sequence of independent random variables, with

$$E[X_i] = \mu_i \quad \text{and} \quad \mathrm{Var}(X_i) = \sigma_i^2 \quad \text{for } i = 1, 2, \ldots$$

Suppose that $0 < \sigma_i^2 \leq M$ for all i. Let a be an arbitrary positive number.

(a) Apply Chebyshev's inequality to show that

$$P\left(\left|\overline{X}_n - \frac{1}{n}\sum_{i=1}^{n}\mu_i\right| > a\right) \leq \frac{\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)}{n^2 a^2}.$$

(b) Conclude from (a) that

$$\lim_{n\to\infty} P\left(\left|\overline{X}_n - \frac{1}{n}\sum_{i=1}^{n}\mu_i\right| > a\right) = 0.$$

And check that the law of large numbers is a special case of this result.

Problem 2. A fair four-sided die is rolled n times. Let

$$S_n = X_1 + X_2 + \cdots + X_n,$$

where X_i is the number showing on the ith roll. We wish to determine a condition on n such that the probability that the absolute difference between the average value S_n/n and the expected value μ_X is less than 20% of μ_X is greater than 92%.

Solve the problem using the Gaussian approximation for S_n, which is suggested by the Central Limit Theorem (CLT). If you need to find $Q(x)$ or $\Phi(x)$, round x to the nearest hundredth.

Note: Do not use the continuity correction for this question, because unless certain values like $2.5n \pm (0.2)n\mu_X$ are integers, inserting the term 0.5 is not applicable.

Problem 3. A factory produces electronic components. The lifetime (in hours) of each component is modeled as a sequence of independent and identically distributed random variables $X_1, X_2, \ldots$, with

$$E[X_i] = 1000, \qquad \mathrm{Var}(X_i) = 400.$$

Let

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

denote the sample mean of the lifetimes of n randomly selected components.

(a) Using the Central Limit Theorem, approximate the probability

$$P\big(\big|\overline{X}_{64} - 1000\big| < 10\big).$$

(b) The factory wants to ensure that

$$P\big(\big|\overline{X}_n - 1000\big| < 10\big) \geq 0.95.$$

Using the Central Limit Theorem, determine the minimum sample size n required to satisfy this condition. Round your answer up to the nearest integer.

Problem 4. A temperature sensor is used to measure the true ambient temperature θ (in degrees Celsius). Due to electronic noise, each measurement is corrupted by random noise.

The statistical model is

$$X_i = \theta + W_i, \quad i = 1, 2, \ldots, n$$

where

- W_i are independent and identically distributed (iid) random variables,
- $E[W_i] = 0$,
- $\mathrm{Var}(W_i) = 9$.

(a) Explain why $X_1, X_2, \ldots, X_n$ form a random sample. State the mean and variance of a single observation X_i.

(b) Define the sample mean

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Compute:

(i) $E[\bar{X}]$,

(ii) $\mathrm{Var}(\bar{X})$.

(c) Let $\hat{\theta} = \bar{X}$ be an estimator of the true temperature θ.

(i) Determine whether $\hat{\theta}$ is an unbiased estimator of θ.

(ii) Compute the bias of $\hat{\theta}$.